

$$\hat{m}c_t = 0$$

$$\Rightarrow \hat{w}_t = \hat{a}_t, \text{ since } \hat{m}c_t = \hat{w}_t - a_t$$

Household intratemporal condition

$$\hat{w}_t = \hat{c}_t + \rho \hat{n}_t$$

$$\Rightarrow \hat{a}_t = \hat{y}_t + \rho(\hat{y}_t - \hat{a}_t)$$

$$\Rightarrow \hat{y}_t = \hat{a}_t$$

Household intertemporal condition

$$r_t^n = \hat{l}_t - E_t \pi_{t+1} = E \hat{c}_{t+1} - \hat{c}_t$$

$$= E \hat{y}_{t+1} - \hat{y}_t$$

$$= E \hat{a}_{t+1} - \hat{a}_t = \rho_a \hat{a}_t - \hat{a}_t = (\rho_a - 1) \hat{a}_t$$